Topology

1. Sets, relations, functions, and order

1.1. Logic

Theorem 1.1.1 (De Morgan's Laws for statements)

AND and OR interchange under complementation.

In C notation:

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!(A || B) == (!A) && (!B)
!(A && B) == (!A) || (!B)
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In mathematical logical notation:

$$\neg(A \vee B) = (\neg A) \wedge (\neg B)$$
$$\neg(A \wedge B) = (\neg A) \vee (\neg B)$$

Theorem 1.1.2 (De Morgan's Laws for set theory)

Union and intersection interchange under complementation.

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

Theorem 1.1.3

Given the true statement $P \Longrightarrow Q$,

- $\neg P \Longrightarrow \neg Q$ is the ***inverse***, which might be true or false.
- $Q \Longrightarrow P$ is the ***converse***, which is true iff the inverse is true.
- $\neg(P \Longrightarrow Q)$ is the ***negation***, which is always false.
- $\neg Q \Longrightarrow \neg P$ is the ***contrapositive***, which is always true.

1.2. Sets

See 01 Set Theory

A set is defined by selecting elements from another set such that $S(x)$ is true:

$$\{x \in A : S(x)\} \text{ or } \{x \in A \mid S(x)\}$$

$\varnothing$ denotes the empty set.

$P \Rightarrow Q$ means P implies Q (if P, then Q).

$P \Leftrightarrow Q$ means P and Q are equivalent (P iff Q).

$A \subset B$ means A is a subset of B (**$A = B$ is allowed**).

$A \subsetneq B$ means A is a strict/proper subset of B.

$A = B$ means $A \subset B$ and $B \supset A$.

$A \cup B$ means the union of A and B.

$A \cap B$ means the intersection of A and B.

$A \parallel B$ means $A \cap B = \varnothing$.

$A \setminus B = \{x \in A : x \notin B\}$, everything in A that's not in B.

$A \times B = \{(a, b) : a \in A, b \in B\}$ is called the ***Cartesian product*** of A and B.

Definition 1.2.1 (Power set)

The set of all subsets of A, denoted $\mathcal{P}(A)$.

1.3. Relations

See 02 Relations

Definition 1.3.1 (Relation)

A ***relation*** $R : A \to B$ is any subset of $A \times B$.

aRb means $(a, b) \in R$.

Theorem 1.3.2 (Relations can compose)

If $R : A \to B$ and $S : B \to C$, then $S \circ R : A \to C$ is a subset of $A \times C$.

⚠ Warning

$R \circ R^{-1}$ is not necessarily the identity map!

Theorem 1.3.3

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$$
$$(T \circ S) \circ R = T \circ (S \circ R)$$

Definition 1.3.4

R is ***reflexive*** iff aRa for all $a \in A$.

R is ***symmetric*** iff $aRb \Leftrightarrow bRa$.

R is ***transitive*** iff $aRb, bRc \Rightarrow aRc$.

1.3.1. Equivalence relations

Definition 1.3.5 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Definition 1.3.6 (Equivalence class)

Let $R : A \to B$ be an equivalence relation. Then the ***equivalence class*** of $a \in A$ is

$$[a] = [a]_R := \{b \in A : aRb\}$$

The set of all equivalence relations of R in A is called A/R.

Lemma 1.3.7

Two equivalence classes are either identical or disjoint.

Definition 1.3.8 (Partition)

A collection of disjoint nonempty subsets of A whose union is all of A.

Theorem 1.3.9

Given an equivalence relation on nonempty A, its set of equivalence classes forms a partition of A.

Given a partition P of A, there is a unique equivalence relation whose set of equivalence classes forms P.

1.3.2. Functions (maps)

See 03 Functions

Definition 1.3.10 (Function)

A ***function*** (also called a ***map*** or ***mapping***) is a relation $f : A \to B$ where each $a \in A$ appears as the first element (input) of an ordered pair $(a, b) = afb$ exactly once.

If $(a, b) \in f$, we can also say $f(a) = b$. b is the ***value*** of f at a, or the ***image*** of a under f.

A is the ***domain*** and B the ***codomain*** of f.

Definition 1.3.11 (Injective)

A function f is ***injective*** or ***one-to-one*** iff:
- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

Definition 1.3.12 (Surjective)

A function $f : A \to B$ is ***surjective*** or ***onto*** iff:
- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

Definition 1.3.13 (Bijective)

A function is ***bijective*** and has ***one-to-one correspondence*** if it is both injective and surjective.

Theorem 1.3.14

A composition of two functions is a function. If both are sur/in/bijective, so is their composition.

Definition 1.3.15

The set of all functions $A \to B$ is denoted B^A.

Lemma 1.3.16

If A and B are finite, then the size of B^A is $|B|^{|A|}$, the size of B to the size of A.

Theorem 1.3.17

For any set A, there is a bijection between $\mathcal{P}(A)$ and 2^A.

Definition 1.3.18 (Induced map)

The ***induced map*** of the relation $f : A \to B$ is a function $\overline{f} : \mathcal{P}(A) \to \mathcal{P}(B)$ defined as

$$\overline{f}(X) \coloneqq \{b \in B : b = f(a) \text{ for some } x \in A\}$$

$\overline{f}(X)$ is called the ***image*** of X. This is sometimes denoted $f(X)$ (which is an abuse of notation).

$\overline{f}$ is always a function regardless of if f is a function.

1.4. Ordering

1.4.1. Partial and total ordering

See 04 Ordering

Definition 1.4.1 (Partial order)

A ***partial order*** $\leq$ on a set A is a relation that is
- **reflexive**: $a \leq a$ for all $a \in A$
- **transitive**: $a \leq b$ and $b \leq c \Rightarrow a \leq c$
- **antisymmetric**: $a \leq b$ and $b \leq a \Rightarrow a = b$

Given a partial order $\leq$, we define the partial order $\geq$ so that $a \geq b \iff b \leq a$.

Definition 1.4.2 (Total order)

A ***total order*** $\leq$ on a set A is a partial order where for all $a, b \in A$, $a \leq b$, $a \geq b$, or both.

Definition 1.4.3 (Strict order)

A ***strict order*** $<$ determined by a partial order $\leq$ is defined s.t. $a < b$ iff $a \leq b$ and $a \neq b$.

Definition 1.4.4 (Partially ordered and totally ordered set)

A ***partially ordered set*** or ***poset*** is a pair of a set and a partial order for that set.

A ***totally ordered set*** is a pair of a set and a total order for that set.

By abuse of notation we also say that the set itself is partially/totally ordered.

Definition 1.4.5 (Upper and lower bounds)

Given a poset A and subset $S \subset A$:

An ***upper bound*** for S is an element $a \in A$ s.t. $s \leq a$ for all $s \in S$.

A ***lower bound*** for S is an element $a \in A$ s.t. $a \leq s$ for all $s \in S$.

The ***supremum*** of S, $\sup S$, is the upper bound such that for any other upper bound u of S, $\sup S \leq u$.

The ***infimum*** of S, $\sup S$, is the lower bound such that for any other lower bound l of S, $l \leq \inf S$.

A set does not have to have a supremum or infimum, but if it does it is unique.

1.4.2. Lattices

See 04 Ordering.2

Definition 1.4.6 (Lattice)

A ***lattice*** is a poset in which every two-element subset has both a supremum and infimum.

A ***complete lattice*** is a poset in which every (including infinite) subset has both a supremum and infimum.

Theorem 1.4.7

The power set of any set, ordered by inclusion, is a complete lattice.

1.4.3. Order-preserving

Definition 1.4.8 (Order-preserving map)

A function $f : A \to B$ between posets is ***order-preserving*** iff whenever $a_1 \leq a_2 \implies f(a_1) \leq f(a_2)$.

Theorem 1.4.9

Every order-preserving map of a complete lattice to itself must have a fixed point (an element of the domain for which $f(a) = a$).

Corollary 1.4.9.1

Let $f : A \to B$ and $g : B \to A$. Then, A and B can be split such that:

- A is split into A_1 and A_2 such that $A_1 \cup A_2 = A$ and $A_1 \cap A_2 = \varnothing$
- B is split A_1 and A_2 such that $A_1 \cup A_2 = A$ and $A_1 \cap A_2 = \varnothing$
- $\overline{f}(A_1) = B_1$ and $\overline{g}(B_2) = A_2$

Definition 1.4.10 (Order isomorphism)

An order-preserving bijection.

1.4.4. Well-ordering

Definition 1.4.11 (Well-ordered set)

A ***well-ordered set*** is a totally-ordered set in which every nonempty subset has a least element.

Equivalently, a set where every element has a "next element" greater than it.

Definition 1.4.12 (Initial segment)

Given poset A, with $a \in A$, the ***initial segment*** of a is the set $\{a' \in A : a' < a\}$.

Lemma 1.4.13

Let f be an order-preserving injection of a well-ordered set into itself. Then $a \leq f(a)$ for all $a \in A$.

A well-ordered set is not order isomorphic to any of its initial segments.

Theorem 1.4.14

Let A or B be well-ordered sets. Then exactly one of the following is true:

1. A is order isomorphic to B.
2. A is order isomorphic to an initial segment of B.
3. B is order isomorphic to an initial segment of A.

Definition 1.4.15 (Least uncountable ordinal)

By the Well-Ordering Theorem, $\mathbb{R}$ has some well-ordering $\mathbb{R}_{\text{wo}}$. Let

$$T := \{a \in \mathbb{R}_{\text{wo}} : a' \text{ is an uncountable initial segment of } \mathbb{R}_{\text{wo}}\}$$

and define $\Omega := \min T$, the ***least uncountable ordinal***.

The ***least uncountable well-ordered set*** is the intial segment of Ω, which satisfies the property that if $a < \Omega$, then the initial segment of a is countable.

1.5. Cardinality

Assume that there is some "overset" containing all of the sets we consider.

Definition 1.5.1 (Cardinality)

$\operatorname{card}(A)$ is the equivalence class of A under bijection

Equivalently, $\operatorname{card}(A) = \operatorname{card}(B)$ iff there is a bijection $A \to B$.

Two sets are ***equipotent*** if they have the same cardinality.

Definition 1.5.2

$\operatorname{card}\ A \leq \operatorname{card}\ B$ iff there is an injection from A to B.

Theorem 1.5.3 (Schroeder-Bernstein Theorem)

$$\operatorname{card}(A) \leq \operatorname{card}\ B \text{ AND } \operatorname{card}(B) \leq \operatorname{card}(A) \implies \operatorname{card}(A) = \operatorname{card}(B)$$

Theorem 1.5.4 (Cantor's Theorem)

For any set A,

$$\operatorname{card}(A) < \operatorname{card}(\mathcal{P}(A))$$

1.6. Countability

Definition 1.6.1 (Countability)

S is countable iff there is an injection $f : S \to \mathbb{Z}^+$, or equivalently iff $\operatorname{card}\ S \leq \operatorname{card}\ \mathbb{Z}^+$.

Corollary 1.6.1.1

1. Every subset of a countable set is countable.

- The elements of a countable set can be indexed by the positive integers.
- All finite sets are countable.

2. A countable union of countable sets is countable.

Theorem 1.6.2

The product of a finite number of countable sets is countable.

However, a product of an infinite number of countable sets is not countable.

Theorem 1.6.3

$$\mathrm{card}(\mathbb{R}) = \mathrm{card}(\mathcal{P}(\mathbb{Z}^+))$$

Thus, the reals are not countable.

1.7. Axiom of choice

Axiom 1.7.1 (Axiom of Choice)

Given a infinite collection of disjoint finite sets, it is possible to select one element from each of the sets.

Equivalently, if A is a set of nomempty sets, there is always a function $f : A \to \cup A$ such that $f(a) \in a$ for all $a \in A$.

In this class, we assume the Axiom of Choice.

Theorem 1.7.2 (Zermelo's Theorem)

The following theorems are equivalent:
- the Axiom of Choice
- the Well-Ordering Property
- the Hausdorff Maximum Principle
- Zorn's Lemma

Theorem 1.7.3 (Well-Ordering Property (Zermelo's Postulate))

Every set can be well-ordered.

Theorem 1.7.4 (Hausdorff Maximum Principle)

Every poset has a "maximal" totally ordered subset, i.e. a totally ordered subset that cannot be extended without escaping the poset.

Theorem 1.7.5 (Zorn's Lemma)

If A is a poset in which every totally ordered subset has an upper bound, then A has a maximal element, i.e. there is no larger element in the poset.

2. Topologies

2.1. Metric spaces

See 06 Metric Spaces

Definition 2.1.1 (Metric)

Given a set M, a ***metric*** $d : M \times M \to \mathbb{R}$ on M is a positive definite symmetric function satisfying the triangle inequality, i.e. it satisfies:
- **Positivity**: $d(x, y) \geq 0$ for all $x, y \in M$
- **Definiteness**: $d(x, y) = 0$ iff $x = y$
- **Symmetry**: $d(x, y) = d(y, x)$
- **Triangle Inequality**; $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in M$

Such a function satisfies everything we expect from a distance.

A ***pseudometric*** satisfies all the above properties except definiteness.

A set equipped with a (pseudo)metric is a ***(pseudo)metric space***.

Theorem 2.1.2 (Metric characterization of continuity)

$f : M \to N$ is continuous at $x \in M$ iff for any $\varepsilon > 0$, there exists $\delta > 0$ s.t. $d(x, y) < \delta \Longrightarrow d(f(x), f(y)) < \varepsilon$.

2.1.1. Epsilon neighborhoods

Definition 2.1.3 (Epsilon neighborhood)

Let N be a metric space.

An ***epsilon neighborhood*** of a set $A \subset N$ is the set

$$U(A, \varepsilon) := \{y \in N : (\exists x \in A : d(x, y) < \varepsilon)\}$$

Equivalently, the set of all points within distance ε of some point in A.

An ***epsilon*** or ***open neighborhood***, ***disk***, or ***ball*** of radius ε around a point x is defined as the set of all points within distance ε of x:

$$U(x, \varepsilon) := \{y \in N : d(x, y) < \varepsilon\} = U(\{x\}, \varepsilon)$$

Theorem 2.1.4 (Epsilon neighborhood definition of continuity)

$f : M \to N$ is continuous at $x \in M$ iff for every open set $V \subset N$ that contains $f(x)$, there is an open set $U \subset M$ containing x s.t. $f(U) \subset V$.

2.1.2. Subspaces

Definition 2.1.5 (Subspace of a metric space)

If $A \subset M$ where d is a metric on M, then $d|_A := d$ restricted to A is a metric on A.

A with this metric is a ***subspace*** of M.

2.1.3. Common metrics

Definition 2.1.6 (Standard metric on $\mathbb{R}^n$)

This is the typical distance on $\mathbb{R}^n$.

$$\mu_2(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Definition 2.1.7 (Taxicab metric on $\mathbb{R}^n$)

Also called the ***Manhattan distance***. It's the distance if you can only travel parallel to one of the axes.

$$\mu_1(x,y) := \sum_{i=1}^{n} |x_i - y_i|$$

Definition 2.1.8 (Chebyshev metric on $\mathbb{R}^n$)

The greatest single difference between any pair of corresponding coordinates of the two points:

$$\mu_\infty(x,y) := \max|x_i - y_i|$$

Definition 2.1.9 (Paris metro metric)

The distance if you can only travel to and from the origin, and all other travel must go through the origin.

$$\mu_{\text{pm}}(x,y) := d(x,0) + d(0,y)$$

Definition 2.1.10 (Discrete metric)

The distance between any two distinct points is 1.

$$\mu_0(x,y) := \begin{cases} 1 \text{ for } x \neq y \\ 0 \text{ for } x = y \end{cases}$$

Definition 2.1.11 (Indiscrete pseudometric)

The distance between everything is 0:

$$\mu_{\text{triv}}(x,y) := 0$$

Also called the ***trivial pseudometric***.

2.2. Topologies

See 07 Topologies

Definition 2.2.1 (Topology)

A ***topology*** $\mathcal{T}$ on the set X is a collection of subsets of X, called the ***open sets***, such that

1. $\mathcal{T}$ is closed under arbitrary unions.
2. $\mathcal{T}$ is closed under finite intersections.

A set together with a topology on it is called a ***topological space***.

Definition 2.2.2 (Closed set)

The complement of an open set.

Lemma 2.2.3

The empty set and whole space are always both closed and open.

Definition 2.2.4 (Metrizable topology)

A topology that is the set of open sets given by some metric.

Definition 2.2.5 (Finer, coarser topologies)

A topology is ***coarser*** (also called ***smaller*** or ***weaker***) than another topology iff it is a subset of the other.

A topology is ***finer*** (also called ***larger*** or ***stronger***) iff it is a superset of the other.

Lemma 2.2.6

The set of all topologies on a specific set is a lattice.

2.3. Closure, interior, boundary

See 07 Topologies.2

Definition 2.3.1 (Closure)

Given $A \subset X$ where X is a topological space, the ***closure*** of A, denoted $\overline{A}$ or $\text{cl } A$, is the intersection of all closed subsets containing A.

Theorem 2.3.2 (Kuratowski closure axioms)

The closure operation satisfies

1. $A \subset \overline{A}$
2. $\overline{\overline{A}} = \overline{A}$
3. $\overline{A \cup B} = \overline{A} \cup \overline{B}$
4. $\overline{\varnothing} = \varnothing$

Definition 2.3.3 (Interior)

Given $A \subset X$ where X is a topological space, the ***interior*** of A, A°, is the union of all open sets contained in A.

Definition 2.3.4 (Boundary)

Given $A \subset X$ where X is a topological space, the ***boundary*** or ***frontier*** of A, denoted ∂A or $\text{bd } A$, is

$$\partial A := \overline{A} - A^\circ$$

Some authors use the word "boundary" to mean only the part of the frontier contained in A.

Definition 2.3.5 (Exterior)

Given $A \subset X$ where X is a topological space, the ***exterior*** is the complement of the closure of A:

$$X - \overline{A} = \overline{A}^c$$

Theorem 2.3.6

Given $A \subset X$ where X is a topological space, X is partitioned into three disjoint parts:

$$X = A^\circ \sqcup \partial A \sqcup X - \overline{A}$$

2.3.1. Bases

See 08 Neighborhoods and Bases.3

Definition 2.3.7 (Base)

Let $\mathcal{T}$ be a topology on X. A ***base*** for $\mathcal{T}$ is a collection of open sets such that any other open set can be made out of unions of the elements of the base.

Definition 2.3.8 (Second countable)

A space is ***second countable*** iff it has a countable base.

Lemma 2.3.9

All second countable spaces are first countable.

Example. Let $\mathcal{T}$ be a countable discrete space. Then the base is at least all the singletons, which is uncountably many. Thus, $\mathcal{T}$ is not second countable.

Definition 2.3.10 (Sub-base)

A collection of sets such that finite intersections of the sets in the sub-base form a base.

Definition 2.3.11 (Topology generated by a sub-base)

The topology generated by a sub-base B is defined such that a set is open iff:

- it is a member of B, or
- it is a finite intersection of members of B, or
- it is an arbitrary union of those intersections

Theorem 2.3.12 (Topology generated by a base)

The topology generated by a base B such that a set is open iff it is an arbitrary union of members of B.

Example. The set of infinite intervals with one open side is a sub-base of $\mathbb{R}$ in the standard topology.

2.4. Neighborhoods

See 08 Neighborhoods and Bases

Definition 2.4.1 (Neighborhood)

Given $A \subset X$ (where X is a topological space), N is a ***neighborhood*** of A iff N contains an open set containing A.

Given $x \in X$, N is a neighborhood of x iff N is a neighborhood of $\{x\}$, i.e. it contains an open set containing x.

$\mathcal{N}(x)$ denotes the set of all neighborhoods of x, which is called the ***neighborhood system*** or ***filter*** for x.

Most of the things we can prove about neighborhoods we can also prove about open neighborhoods, so some authors define neighborhoods to always be open.

Lemma 2.4.2

$U \subset X$ is open iff for each $x \in U$ there is a neighborhood N_x of x contained in U.

Axiom 2.4.3 (Hausdorff neighborhood axioms)

Let $x \in X$. Then,

1. x is in every one of its neighborhoods:
$$N \in \mathcal{N}(x) \Longrightarrow x \in N$$
2. $\mathcal{N}(x)$ is closed under supersets:
$$N \in \mathcal{N}(x), N \subset M \Longrightarrow M \subset \mathcal{N}(x)$$
3. $\mathcal{N}(x)$ is closed under finite intersection.
4. For each neighborhood N of x, it contains a neighborhood N' such that for each item y in N', N is a neighborhood of y:
$$\begin{gathered} N \in \mathcal{N}(x) \Longrightarrow (\exists N' \in \mathcal{N}(x) : \\ N' \subset N \text{ and } y \in N' \Longrightarrow N \subset \mathcal{N}(y)) \end{gathered}$$

Conversely, if each $x \in X$ has a collection of neighborhoods $\mathcal{N}'(x)$ satisfying the above axioms, than there is a unique topology on X for which $\mathcal{N}'(x) = \mathcal{N}(x)$ for all $x \in X$.

Theorem 2.4.4 (Neighborhood characterization of closure)

Let $x \in X$, $A \subset X$. Then $x \in \overline{A}$ iff every neighborhood of x meets A.

Theorem 2.4.5 (Neighborhood characterization of boundary)

∂A is the set of all points $x \in X$ s.t. every neighborhood of x meets both A and $X - A$.

2.4.1. Neighborhood bases

Definition 2.4.6 (Neighborhood base)

Let $x \in X$. A ***neighborhood base*** of x is a set $\mathcal{B}(x) \subset \mathcal{N}(x)$ such that for any $N \in \mathcal{N}(x)$ there is a $B \in B(x)$ such that $B \subset N$.

Equivalently, it is a set of ***basic neighborhoods*** around a x s.t. each neighborhood of x contains a basic neighborhood.

Equivalently, it is a filter base of the neighborhood filter $\mathcal{N}$.

Definition 2.4.7 (First countable)

A space is ***first countable*** iff if each point has a countable neighborhood base.

Example. Let $\overline{\Omega}$ be the minimal uncountable well-ordered set with its endpoint ω_0. $\mathcal{B}(\omega_0)$ is not countable, so $\overline{\Omega}$ is not first countable.

Example. Every metrizable space is first countable.

Axiom 2.4.8 (Neighborhood axioms)

Let $\mathcal{B}(x)$ be a neighborhood base at x. Then

1. There exist basic neighborhoods of x.
2. x is contained in each of its basic neighborhoods:
$$U \in \mathcal{B}(x) \Longrightarrow x \in U$$

3. $\mathcal{B}(x)$ is closed under finite intersection:

$$U_1, U_2 \in \mathcal{B}(x) \implies (\exists V \in \mathcal{B}(x) : V \subset U_1 \cap U_2)$$

4. For each basic neighborhood U of x, it contains a basic neighborhood V such that for each item y in V, V contains a basic neighborhood W of y:

$$U \in \mathcal{B}(x) \implies (\exists V \in \mathcal{B}(x) :$$
$$V \subset U \text{ and } (y \in V \implies \exists W \in \mathcal{B}(y) : W \subset V))$$

Conversely, given a set $\mathcal{B}(x)$ for each $x \in X$ satisfying the above axioms, there is a unique topology $\mathcal{T}$ on X s.t. for all $x \in X$, $\mathcal{B}(x)$ is a neighborhood base of x in the topology $\mathcal{T}$. $\mathcal{T}$ is defined by:

$$\mathcal{T} := \{U \subset X : x \in U \implies (\exists V \in \mathcal{B}(x) \ni V \subset U)\}$$

2.4.2. Accumulation points

Definition 2.4.9 (Accumulation point)

x is an ***accumulation point*** (also called ***limit point*** or ***cluster point***) of A iff every neighborhood of x contains a point of A other than x.

Definition 2.4.10 (Derived set)

The set of accumulation points of A is called the ***derived set***, denoted A'.

Corollary 2.4.10.1 (Accumulation point characterization of closure)

$\overline{A} = A \cup A'$.

2.5. Continuity

See 09 Continuous Functions

Definition 2.5.1 (Continuous function)

$f : X \to Y$ is a continuous function at $x_0 \in X$ iff $f^{-1}(N)$ is a neighborhood of x_0 for every neighborhood N of $f(x_0)$.

Equivalently, it is continuous iff every neighborhood of $f(x)$ pulls back (by f^{-1}) to a neighborhood of x.

f is continuous if it is continuous at all $x_0 \in X$.

Definition 2.5.2 (Homeomorphism)

f is a homeomorphism iff f is a continuous bijection whose inverse is also continuous.

Homeomorphisms are equivalence relations between topological spaces.

Theorem 2.5.3 (Continuity composes)

If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f : X \to Z$ is continuous at x_0.

Theorem 2.5.4

Given $f : X \to Y$, the following are equivalent.

1. f is continuous.
2. $f\left(\overline{A}\right) \subset \overline{f(A)}$ for all $A \subset X$.
3. If $B \subset Y$ is closed, then $f^{-1}(B)$ is closed in X.
4. If $B \subset Y$ is open, then $f^{-1}(B)$ is open in X.

Lemma 2.5.5

$f : X \to Y$ is continuous iff $f^{-1}(c)$ is open for each c in some subbase of Y.

Lemma 2.5.6

Any function from any discrete space to any other space is continuous.

Any function from any space to any indiscrete space is continuous.

2.6. Initial (induced) topology

See 10 Induced Topologies

2.6.1. Subspace topology

Definition 2.6.1 (Subspace topology)

Given $A \subset X$, where X is a topological space, the ***subspace topology*** (***relative topology***) on A is defined such that $B \subset A$ is open in A iff $B = A \cap O$ for some open set O in X.

The subspace topology is the initial topology induced by the inclusion map $i_A : A \to X$ defined by $i_A(x) := x$.

A equipped with the subspace topology is called a ***subspace*** of X.

Example. $\mathbb{Q}^\circ$ under the standard topology on $\mathbb{R}$ is empty, since there are no open intervals in $\mathbb{Q}$.

However, $\mathbb{Q}^\circ$ under the subspace topology on $\mathbb{Q}$ is the entire space, since all spaces are open.

Lemma 2.6.2

The following properties are preserved by taking subspaces:

- discrete
- indiscrete
- metrizable
- subspace of some space

2.6.2. Initial (induced) topology

Definition 2.6.3 (Initial topology)

For each α in some index set $\mathcal{A}$, let A_α be a topological space and f_α be a function $X \to A_\alpha$. Then, the initial topology ***induced*** on X by the set of f_α is the coarsest topology in X for which each f_α is continuous.

Equivalently, it is the intersection of all of the topologies for which each f_α is continuous.

Also called: ***initial topology***, ***strong topology***, ***limit topology***, ***projective topology***

Equivalently, given a bunch of sets A_α and functions $f_\alpha : X \to A_\alpha$, pull the open sets in each A_α back to X (by f_α), and define these pullbacks to be open on X. This is a sub-base; generate a topology from this sub-base by defining the finite intersections and arbitrary to be open.. This forms a topology on X, which is called the ***initial topology***.

Lemma 2.6.4

The initial topology on X induced by $\{f_\alpha\}$ is the topology generated by the following sub-base: the collection of all $f_\alpha^{-1}(U)$ for all open sets $U \subset X_\alpha$.

Theorem 2.6.5 (Universal property of the initial topology)

The initial topology induced by $\{f_\alpha\}$ is the unique topology where if Y is any topological space, then $g : Y \to X$ is continuous iff each $g \circ f_\alpha : Y \to A_\alpha$ is continuous:

$$\begin{tikzcd}
(Y, \mathcal{T}) \arrow[d, "g"] \arrow[dr, "f_\alpha \circ g"] & \\
(X, \mathcal{T}_{\text{initial}}) \arrow[r, "f_\alpha"] & (X_\alpha, \mathcal{T}_\alpha)
\end{tikzcd}$$

Definition 2.6.6 (Open and closed functions)

Let X and Y be topological spaces. $f : X \to Y$ is ***open*** iff $f(U)$ is open for all U open on X.

$f : X \to Y$ is ***closed*** iff $f(C)$ is closed for all C closed on X.

(These are not opposite.)

2.6.3. Product topology

See Munkres 2§19 (p.113)

Definition 2.6.7 (Projection)

Let $\{X_\alpha\}$ be a collection of sets, and $X = \prod_{\alpha \in A} X_\alpha$. Then for each $\alpha \in A$, the ***projection map*** $\pi_\alpha : X \to X_\alpha$ maps each tuple in A to its αth element.

Definition 2.6.8 (Product topology)

Let $\{X_\alpha\}$ be a set of topological spaces, and $X = \prod_{\alpha \in A} X_\alpha$. The topology on X induced by the collection of projections $\{\pi_\alpha : \alpha \in A\}$ is called the ***product topology***.

Equivalently, given a product space X, pull the open sets in each X_α back to X (using the projection maps π_α), and define these pullbacks to be open on X. This is a sub-base; generate a topology from this sub-base by defining the finite intersections and arbitrary unions to be open. This forms a topology on X, which is called the ***product topology***.

Lemma 2.6.9 (Product topology constrains finitely many axes)

The collection of sets which are products of open sets in each of the X_α, where all but a finite number are the entire set X_α, form a base for the product topology.

Thus, only a finite number of axes can be constrained when creating an open set in the product topology.

This is because the pullbacks of open sets in one of the X_α only constrain a single axis, and we can only take finite intersections of them.

Theorem 2.6.10

If $\mathcal{A}' \subset \mathcal{A}$, then $\pi : \prod_{a \in \mathcal{A}} X_\alpha \to \prod_{a \in \mathcal{A}'} X_\alpha$ is continuous and open.

2.6.4. Box topology

Definition 2.6.11 (Box topology)

Let $\{X_\alpha\}$ be a set of topological spaces, and $X = \prod_{\alpha \in A} X_\alpha$. The ***box topology*** on X is generated by the base

$$\left\{ \prod_{\alpha \in A} U_\alpha : U_\alpha \text{ open in } X_\alpha \right\}$$

2.7. Covers and defining families

See 10 Induced Topologies.3

Definition 2.7.1 (Defining family)

A ***defining family*** for topological space X is a collection $\mathcal{C}$ s.t. $U \subset X$ is open in X iff its intersection with each $C \in \mathcal{C}$ is open in the subspace topology on C.

Definition 2.7.2 (Open cover)

An ***open cover*** of X is a collection of open sets whose union contains all of X.

All open covers are defining families.

Definition 2.7.3 (Locally finite)

A collection of subsets of a topological space is ***locally finite*** if each point has a neighborhood that meets only finitely many sets in the collection.

Definition 2.7.4 (Closed cover)

An ***open cover*** of X is a collection of closed sets whose union contains all of X.

Locally finite closed covers are defining families.

Theorem 2.7.5

Let $\mathcal{C}$ be a defining family for X. Then $f : X \to Y$ is continuous iff its restriction to each $C \in \mathcal{C}$ is continuous.

2.8. Final (coinduced) topologies

Definition 2.8.1 (Final (coinduced) topology)

For each α in some index set $\mathcal{A}$, let A_α be a topological space and g_α be a function $A_\alpha \to Y$. Then, the final topology ***coinduced*** on Y by the set of g_α is the finest topology in Y for which each g_α is continuous.

Equivalently, it is the union of all of the topologies for which each g_α is continuous.

Also called: ***final topology***, ***weak topology***, ***colimit topology***, ***inductive topology***

Equivalently, a set U is open in the final topology iff its preimage $g_\alpha^{-1}(U)$ is open in the topology on A_α for every α.

Theorem 2.8.2 (Universal property of the final topology)

The final topology induced by $\{g_\alpha\}$ is the unique topology where $f : Y \to Z$ is continuous iff $f \circ g_\alpha : X_\alpha \to Z$ is continuous for every α.

$$\begin{tikzcd}
(Z, \mathcal{T}) & \\
(Y, \mathcal{T}_{\text{final}}) \arrow[u, "f"] & (X_\alpha, \mathcal{T}_\alpha) \arrow[l, "g_\alpha"] \arrow[ul, "f \circ g_\alpha"']
\end{tikzcd}$$

2.8.1. Quotient topology

Definition 2.8.3 (Quotient topology)

Given topological space X and surjection $g : X \to Y$, the final topology coinduced by g is the ***quotient topology***.

Then, Y is the ***quotient space*** of X by g, and is considered one of the ***identification spaces*** of X.

Definition 2.8.4 ($X/\sim$)

Let X be a space and $\sim$ be an equivalence relation, with Y being its set of equivalence classes. The space $\boldsymbol{X/\sim}$ is defined to be the quotient topology of X coinduced by $g : X \to Y$ defined as

$$g(x) = [x]_\sim \text{ (the equivalence class of } x\text{)}$$

2.8.2. Topological sum

Definition 2.8.5 (Topological sum)

Let $\{Y_\alpha\}$ be a collection of disjoint sets. Then, $X = \bigsqcup Y_\alpha$ is the ***topological sum*** of the Y_α, denoted $\bigoplus Y_\alpha$.

The topology on $\bigoplus Y_\alpha$ is such that a set is open iff it is open in one of the Y_α, or it is a union of open sets in different Y_α.

Such a space is obviously disconnected.

This topology is equivalent to the final topology coinduced by the inclusion maps $i_\alpha : Y_\alpha \to \bigsqcup Y_\alpha$.

Theorem 2.8.6 (Scissors and Paste Theorem)

Let $\mathcal{C}$ be a defining family for topological space X that covers X. Let Y be the topological sum of all the $C \in \mathcal{C}$. Define $f : Y \to X$ as the identity map $f(y) = y$. Then X is the quotient space of Y by f.

3. Limits

3.1. Sequences

Definition 3.1.1 (Sequence)

A function $x : \mathbb{N} \to X$, where X is a topological space. This is denoted $\{x_n\}_{n=1}^\infty$, where $x_n = x(n) \in X$.

Definition 3.1.2 (Convergence of a sequence)

x_n converges to x_0 ($x_n \to x_0$) iff for every neighborhood U around x_0, there exists N_U s.t. $n > N_U$ implies $x_n \in U$. Equivalently, every neighborhood x_0 contains all but finitely many of the x_n.

Lemma 3.1.3

In a metric space, $x_n \to x_0$ iff

$$\lim_{n\to\infty} d(x_n, x_0) = 0$$

Lemma 3.1.4

In the product space $X = \prod X_\alpha$, $x_n \to x_0$ in X iff $\pi_\alpha(x_n) \to \pi_\alpha(x_0)$ for every α.

Theorem 3.1.5

In a first countable space, $x \in \overline{A}$ iff there is a sequence of points in A that converge to x.

Theorem 3.1.6

If $f : X \to Y$ is continuous at x_0 and $\{x_n\} \to x_0$, then $f(x_n) \to f(x_0)$.

Conversely, if x_0 has a countable neighborhood base and each sequence $\{x_n\} \to x_0$, $f(x_n) \to f(x_0)$, then f_0 is continuous at x_0.

3.2. Nets

Definition 3.2.1 (Directed upward)

A poset in which any two elements have an upper bound. That is, given $\alpha, \beta \in S$, there exists $\gamma \in S$ s.t. $a \leq \gamma$ and $\beta \leq \gamma$

Definition 3.2.2 (Net)

A subset of a topological space which is directed upward by some $\leq$ (which is not necessarily related to the space.)

This is sometimes written using a directed-upward set $\mathcal{A}$, where the net is denoted $\{x_\alpha\}_{\alpha \in \mathcal{A}}$

Definition 3.2.3 (Convergence of a net)

$x_\alpha \to x_0$, where $\{x_\alpha\}_{\alpha \in \mathcal{A}}$ is a net, iff for each neighborhood N around x_0, there exists $\gamma \in \mathcal{A}$ s.t. $x_\alpha \in N$ for all $\alpha \geq \gamma$.

Lemma 3.2.4 (Net characterization of closure)

If A is a subset of a topological space, $x_0 \in \overline{A}$ iff there exists a net in A converging to x_0.

Lemma 3.2.5 (Net characterization of continuity)

$f : X \to Y$ is continuous at x_0 iff for each net $x_\alpha \to x_0$, $f(x_\alpha) \to f(x_0)$.

3.3. Filters

Definition 3.3.1 (Filter base)

A ***filter base*** $\mathcal{F}$ in the set X is a nonempty collection of nonempty subsets of X which is directed downward by inclusion.

Definition 3.3.2 (Filter)

A filter base $\mathcal{F}$ which is closed under supersets:

$$\forall A \in \mathcal{F} \quad \nexists B \in X : A \subsetneq B$$

Lemma 3.3.3

The collection of all sets in a filter base together with their supersets is a filter.

Definition 3.3.4 (Canonical filter)

The ***canonical filter generated by*** $\boldsymbol{A}$ is the set of all supersets of A, if A is nonempty.

Definition 3.3.5 (Fréchet filter)

The collection of subsets of the nonnegative integers whose complements are finite.

Lemma 3.3.6 (Images of filter bases are filter bases)

If $\mathcal{F}$ is a filter base on X, and $f : X \to Y$, then

$$f(\mathcal{F}) = \{f(F) : F \in \mathcal{F}\}$$

is a filter base on Y.

Definition 3.3.7 (Elementary filter)

The image of the Fréchet filter.

Definition 3.3.8 (Convergence of a filter base)

$\mathcal{F} \to x_0$ iff every neighborhood N of x_0 contains some $F \subset \mathcal{F}$.

Lemma 3.3.9 (Constructing filter base from net)

Given the nonempty net $\{x_\alpha\}_{\alpha \in \mathcal{A}}$ in X, we can define

$$X_\alpha = \{x_{\alpha'} : \alpha' < \alpha\}$$

Then, $\{X_\alpha\}$ is a filter base in X, and $x_\alpha \to x_0$ iff $X_\alpha \to x_0$.

Lemma 3.3.10 (Collection of neighborhoods is filter)

The collection of all neighborhoods of any nonempty set is a filter.

Lemma 3.3.11 (Filter characterization of continuity)

Let $f : X \to Y$, and $x_0 \in X$ where N is the set of neighborhoods of x_0. Then, f is continuous at x_0 iff $N \to f(x_0)$.

Lemma 3.3.12 (Filter base characterization of closure)

$x \in \overline{A}$ iff there is a filter base in A converging to x.

Theorem 3.3.13

Given a filter base $\mathcal{F}$ in the product space $X = \prod X_\alpha$, $\mathcal{F} \to x_0$ iff each $\pi_{\alpha(\mathcal{F})} \to \pi_\alpha(x_0)$ in X_α.

3.3.1. Ultrafilters

Lemma 3.3.14

The set of all filter bases on a specific set is a poset under inclusion.

The set of all filter bases containing a given filter base is a poset where every totally ordered subset has an upper bound.

Definition 3.3.15 (Ultrafilter)

The maximal filter base in a specific set; i.e. the filter base to which no more sets can be added while maintaining downward direction.

The ultrafilter is a filter.

Theorem 3.3.16 (Test for ultrafilter)

A filter base $\mathcal{F}$ is an ultrafilter iff for each $E \subset X$ either $E \in \mathcal{F}$ or $X - E \in \mathcal{F}$.

Definition 3.3.17 (Cluster point)

A ***cluster point*** of a filter base $\mathcal{F}$ is a point that is in $\overline{F}$ for every $F \in \mathcal{F}$.

Theorem 3.3.18

x is a cluster point of $\mathcal{F}$ iff there exists an ultrafilter $\mathcal{F}'$ containing $\mathcal{F}$ that converges to x.

4. Separation

4.1. Separated

Definition 4.1.1 (Separated)

A and B in the space X are ***separated*** iff

$$A \cap \overline{B} = \overline{A} \cap B = \varnothing$$

or equivalently iff there exist neighborhoods of each that do not include the other.

Definition 4.1.2 (Separated by neighborhoods)

A and B in the space X are ***separated by neighborhoods*** iff there exist neighborhoods U of A and V of B s.t. $U \cap V = \varnothing$.

Being separated by neighborhoods is equivalent to being separated by open neighborhoods.

Separated by neighborhoods implies separated.

Definition 4.1.3 (Separated by closed neighborhoods)

A and B in the space X are ***separated by closed neighborhoods*** iff there exist neighborhoods U of A and V of B s.t. $\overline{U} \cap \overline{V} = \varnothing$.

Separated by closed neighborhoods implies separated by neighborhoods.

Definition 4.1.4 (Separated by continuous functions)

A and B in the space X are ***separated by continuous functions*** iff there exists a continuous function such that $f(x) = 0$ for all $x \in A$ and $f(x) = 1$ for all $x \in B$.

Separated by continuous functions implies separated by neighborhoods.

4.2. T_0 through T_2 spaces

Definition 4.2.1 (Kolmogorov space (T_0))

A space where given two distinct points in the space, at least one has a neighborhood that does not include the other.

Lemma 4.2.2

X is T_0 iff for any $x, y \in X$, $x \neq y \Leftrightarrow \overline{\{x\}} \neq \overline{\{y\}}$.

Definition 4.2.3 (Fréchet space (T_1))

A space where given two distinct points in the space, each has a neighborhood that does not include the other.

Equivalently, a space where all two points are separated.

T_1 spaces are T_0.

Lemma 4.2.4

Points in T_1 spaces are closed sets.

Lemma 4.2.5

X is T_1 iff $\{x\} = \overline{\{x\}}$ for all $x \in X$.

Definition 4.2.6 (Hausdorff space (T_2))

A space where given two distinct points $x \neq y$, there are neighborhoods X of x and Y of y s.t. $X \cap Y = \varnothing$.

Equivalently, a space where all two points are separated by neighborhoods.

T_2 spaces are T_1.

Definition 4.2.7 ($T_{2\frac{1}{2}}$ space)

A space where given two distinct points $x \neq y$, there are neighborhoods X of x and Y of y s.t. $\overline{U} \cap \overline{V} = \varnothing$.

Equivalently, a space where all two points are separated by closed neighborhoods.

This is also called ***completely Hausdorff*** (in older texts) and ***Urysohn*** (in newer texts).

$T_{2\frac{1}{2}}$ spaces are T_2.

Definition 4.2.8 (Functionally Hausdorff space)

A space X where given two points $x \neq y$, there is a continuous function $f : X \to [0, 1]$ with $f(x) = 0$ and $f(y) = 1$.

Equivalently, a space where all two points are separated by continuous functions.

This is also called ***Urysohn*** (in older texts) and ***completely Hausdorff*** (in newer texts).

Functionally Hausdorff spaces are $T_{2\frac{1}{2}}$.

Lemma 4.2.9

The following are equivalent:
1. X is T_2
2. Limits of filter bases in X are unique.
3. The diagonal, $\Delta = \{(x, y) \in X \times X : x = y\}$, is a closed set in the product space.

Lemma 4.2.10

Products and subspaces of T_i spaces are T_i for $i = 0, 1, 2$.

4.3. Regular spaces

Definition 4.3.1 (Regular space)

A space in which any point and any disjoint closed set can be separated by neighborhoods.

Equivalently a space in which any point and any disjoint closed set can be separated by closed neighborhood.

Theorem 4.3.2

X is regular iff the closed neighborhoods of each $x \in X$ form a neighborhood base of x.

Definition 4.3.3 (Regular Hausdorff space (T_3))

A T_1 space which is regular.

Equivalently, a $T_{2\frac{1}{2}}$ space which is regular.

Definition 4.3.4 (Completely regular space)

A space in which closed sets and disjoint single points can be separated by continuous functions.

Definition 4.3.5 (Tychonoff space ($T_{3\frac{1}{2}}$))

A completely regular T_1 space.

Tychonoff spaces are T_3 and functionally Hausdorff.

Theorem 4.3.6

Subspaces and products of regular spaces are regular.

Subspaces and products of completely regular spaces are completely regular.

Theorem 4.3.7

X is completely regular iff it has the topology induced by its set of bounded continuous real-valued functions.

Corollary 4.3.7.1

A topological space is Tychonoff iff it is homeomorphic to a subspace of a hypercube ($[0, 1]^n$).

4.4. Normal spaces

Definition 4.4.1 (Normal space)

A topological space is ***normal*** iff disjoint closed sets are separated by neighborhoods.

Definition 4.4.2 (Normal Tychonoff space (T_4))

A normal T_1 space.

Equivalently, a normal $T_{3\frac{1}{2}}$ space.

Lemma 4.4.3

If X contains a dense set D and a closed discrete subspace S with cardinality at least as large as $\mathcal{P}(D)$, then X cannot be normal.

Definition 4.4.4 (Completely normal)

A space in which every subspace is normal.

Definition 4.4.5 (Completely normal Tychonoff (T_5))

A completely normal T_1 space.

Equivalently, a completely normal T_4 space.

Theorem 4.4.6 (Urysohn's Lemma)

In a normal space X, any two disjoint closed sets A and B can be separated by a continuous function.

Theorem 4.4.7 (Tietze Extension Theorem)

X is normal iff for any closed set A of X, every continuous function $f : A \to [0, 1]$ can be extended to a continuous function $X \to \mathbb{R}$.

Corollary 4.4.7.1

X is normal iff for any closed set A of X, every continuous function $f : A \to \mathbb{R}$ can be extended to a continuous function $X \to \mathbb{R}$.

5. Countability properties

See 17 Countability Properties

Definition 2.4.7 (First countable)

A space is ***first countable*** iff if each point has a countable neighborhood base.

Definition 2.3.8 (Second countable)

A space is ***second countable*** iff it has a countable base.

Definition 5.1 (Separable)

A space is ***separable*** iff it has a countable dense subset, i.e. it has a countable subset s.t. every nonempty open set of X intersects the countable subset, or equivalently the closure of the countable subset is the whole space.

Lemma 5.2

Second countability is preserved under subspaces and countable products.

Definition 5.3 (Lindelöf space)

A space is ***Lindelöf*** iff every open cover has a countable subcover.

Lemma 5.4

Second countability implies first countability, separability, and Lindelöf.

Lemma 5.5

Any collection $\mathcal{U}$ of open subsets of a second countable space has a countable subcollection $\mathcal{V}$ s.t.

$$\bigcup_{V \in \mathcal{V}} V = \bigcup_{U \in \mathcal{U}} U$$

Thus, every open cover of a second countable space has a countable subcover, so all second countable spaces are Lindelöf.

Lemma 5.6

All regular Lindelöf spaces are normal.

In metric and pseudometric spaces, separability, Lindelöf, and second countability are equivalent.

Definition 5.7 (Separation)

Given $A \subset X$, two sets B and C are a ***separation*** of A iff they are nonempty and

$$B \cap \overline{C} = \varnothing = \overline{B} \cap C$$

Theorem 5.8

A subspace $A \subset X$ is disconnected iff there exists a separation of it.

6. Connectedness

See 18 Connectedness

Definition 6.1 (Connected space)

A topological space where the only sets that are both open and closed are the whole space and $\varnothing$.

Theorem 6.2

The continuous image of a connected space is connected.

Definition 6.3 (Separation)

A ***separation of A*** is a pair B, C of nonempty subsets of A which satisfies

$$B \cap \overline{C} = \overline{B} \cap C = \varnothing$$

Theorem 6.4

A subspace A of X is disconnected under the subspace topology iff it has a separation.

Corollary 6.4.1

If A is connected and $A \subset B \cup C$, where $B \cap \overline{C}$ and $\overline{B} \cap C$ are both empty, though neither B nor C are, then either $A \subset B$ or $A \subset C$.

If A is a connected subset of a connected space, and $X - A = B \cup C$ where B and C are separated, then $A \cup C$ and $B \cup C$ are both connected.

If A and B are closed sets s.t. both $A \cap B$ and $A \cup B$ are connected, then A and B are both connected.

If A is connected and $A \subset B \subset \overline{A}$, then B is connected.

Let $\mathcal{A}$ be any collection of connected subsets of X with the property that any two elements of $\mathcal{A}$ have nonempty intersections. Then $\mathcal{A} = \cup_{A \in \mathcal{A}} A$ is connected.

X is connected iff for each pair of points, there is a connected subset of X containing both of them.

Theorem 6.5

The connected subsets of $\mathbb{R}$ are the intervals (a, b), $(a, b]$, $[a, b)$, and $[a, b]$ where a and b are real numbers or infinite.

Theorem 6.6 (Intermediate Value Theorem)

If $f : X \to \mathbb{R}$ is continuous with X connected and $x, y \in X$, then for any $c \in (f(a), f(b))$ there exists $z \in X$ s.t. $f(z) = c$.

Theorem 6.7

A product of connected spaces is connected. Conversely, given a nonempty product that is connected, each factor is connected.

6.1. Components

Definition 6.1.1 (Component)

A connected subset of X not contained in any other connected subset of X.

Lemma 6.1.2

The union of all connected subsets of X containing x is a component containing x.

Lemma 6.1.3

The components of a space form a partition of the space.

Lemma 6.1.4

The components of a product space are the products of the components of the factors.

Definition 6.1.5 (Totally disconnected space)

A space in which each component is a single point.

Lemma 6.1.6

Products of totally disconnected spaces are totally disconnected.

Discrete spaces and the Cantor set are both totally disconnected.

6.2. Local connectedness

Definition 6.2.1 (Locally connected space)

A space in which each point has a neighborhood base consisting of connected sets.

Theorem 6.2.2

The **finite** product of locally connected sets are locally connected. If a product is locally connected, then all but finitely many factors are connected and the remainder locally connected.

Theorem 6.2.3

The following are equivalent:
1. X is locally connected.
2. If N is any neighborhood of x there is a neighborhood M of x contained in the component of N that contains x.
3. Components of open sets are open.

7. Compactness

See 19.1 Types of Compactness

Definition 7.1 (Compactness)

A space where every open cover has a finite subcover.

Definition 7.2 (Countable compactness)

A space where every countable cover has a finite subcover.

Lemma 7.3

A space is compact iff it is countably compact and Lindelöf.

Definition 7.4 (Sequential compactness)

A space where every sequence has a convergent subsequence.

Lemma 7.5

Sequential compactness implies countable compactness.

Example. Closed, bounded intervals on the real line are compact and sequentially compact.

The reals are neither compact nor sequentially compact.

$[0, \Omega]$ is both compact and sequentially compact.

$[0, \Lambda)$ for any infinite Λ is not compact, but $[0, \Omega)$ is sequentially compact.

Infinite discrete spaces are neither compact nor sequentially compact.

Definition 7.6 (Finite intersection property)

A collection of subsets of X has the ***finite intersection property*** iff every finite subcollection of the sets has a nonempty intersection.

If X is a topological space, this is equivalent to being a filter base.

Theorem 7.7

The following are equivalent:
1. X is compact.
2. Every family of closed subsets of X with the finite intersection property has nonempty intersection.
3. Every ultrafilter in X has a limit.

Corollary 7.7.1 (Tychonoff Theorem)

A product $X \prod X_\alpha$ of nonempty spaces is compact iff each factor is compact.

Theorem 7.8

In a T_1 space, the following are equivalent:
1. X is countably compact
2. Every infinite open cover has a proper subcover
3. Every infinite subset has an accumulation point
4. (***Cantor Intersection Property***) Every decreasing sequence of nonempty closed subsets has nonempty intersection.

Theorem 7.9

If X is countably compact, every infinite open cover has a proper subcover.

If every infinite open cover has a proper subcover, every infinite subset has an accumulation point.

7.1. Separation properties in compact spaces

See 19b Separation Properties in Compact Spaces

Lemma 7.1.1

Let A be a compact subset of a Hausdorff space X with $x \notin A$. Then, x and A can be separated by open sets.

Corollary 7.1.1.1

Compact subsets of Hausdorff spaces are closed.

Corollary 7.1.1.2

A continuous bijection from a compact space to a Hausdorff space is a homeomorphism.

Lemma 7.1.2

In a (completely) regular space, a compact set and a disjoint closed set can be separated by neighborhoods (continuous functions).

Corollary 7.1.2.1

All compact Hausdorff spaces are T_4.

7.2. Local compactness

See 20 Local Compactness

Definition 7.2.1 (Local compactness)

A space X is ***locally compact*** iff each point $x \in X$ has a neighborhood base consisting of compact neighborhoods.

Lemma 7.2.2

All locally compact Hausdorff spaces are T_3.

Lemma 7.2.3

A Hausdorff space is locally compact iff each point has at least one compact neighborhood.

Corollary 7.2.3.1

Compact Hausdorff spaces are locally compact.

Theorem 7.2.4

A subset of a locally compact Hausdorff space is itself locally compact iff it is the intersection of a closed set with an open set.

7.3. One-point compactification

Definition 7.3.1 (One-point compactification)

Let X be a space, and define ∞_X to be some point not in X. Create a new space, called the ***one-point compactification*** or ***Alexandroff compactification*** of X, $X^* := X \cup \{\infty_X\}$, whose open sets are the open sets of X, together with complements of closed compact subsets of X in X^* (which all contain ∞_X).

This forms a closed compact space.

Lemma 7.3.2

The one-point compactification of $\mathbb{R}^n$ is $\mathbb{S}^n$, the n-dimensional sphere.

The one-point compactification of an open interval on $\mathbb{R}$ is $\mathbb{S}^1$, a circle.

The one-point compactification of $[a, b)$ in the real line is $[a, b]$.

7.4. Stone-Čech compactification

Definition 7.4.1 (Stone-Čech compactification)

A ***Stone-Čech compactification*** of a topological space X is a compact Hausdorff space $\beta(X)$ together with an embedding $e : X \to \beta(X)$ such that every bounded real-valued continuous function on X has a unique continuous extension to $\beta(X)$. (An ***embedding*** is a homeomorphism of X with a subspace of $\beta(X)$.)

Theorem 7.4.2

X has a Stone-Čech compactification iff X is Tychonoff ($T_{3\frac{1}{2}}$).

Theorem 7.4.3

Let $e : X \to \beta(X)$ be a Stone-Čech compactification of X and let $f : X \to Y$ be any continuous map from X to a compact Hausdorff space. Then, f has a unique continuous extension to all of $\beta(X)$, i.e. there exists a function $F : \beta(X) \to Y$ s.t. $F \circ e = f$.

Corollary 7.4.3.1

All Stone-Čech compactifications of X are homeomorphic.

7.5. Paracompactness

Definition 7.5.1 (Refinement)

A collection of sets $\mathcal{V}$ is a ***refinement*** of $\mathcal{U}$ iff each $V \in \mathcal{V}$ is a subset of some $U \in \mathcal{U}$.

Definition 7.5.2 (Paracompact)

A space is ***paracompact*** iff every open covering has a refinement that is a locally finite open cover.

Lemma 7.5.3

The following spaces are paracompact:
- $\mathbb{R}$
- The line with two origins
  - but note that it is not Hausdorff

Lemma 7.5.4

Every compact space is paracompact.

Definition 7.5.5 (Countably locally finite)

A collection of sets is ***countably locally finite*** iff it is a countable union of locally finite collections.

Theorem 7.5.6

For a T_3 space X, the following are equivalent:
1. X is paracompact.
2. Every open covering of X has a countable locally finite open refinement which covers X.
3. Every open covering of X has a locally finite refinement that covers X (the sets need not be open).
4. Every open covering of X has a locally finite closed refinement that covers X.

Corollary 7.5.6.1

A Lindelöf T_3 space is paracompact.

Lemma 7.5.7

Let A be a closed subset of a paracompact space. Let $\mathcal{U}$ be a collection of open subsets of X covering A. Then, there is a locally finite open refinement of $\mathcal{U}$ that covers A.

Corollary 7.5.7.1

A closed subset of a paracompact space is paracompact.

Theorem 7.5.8

A paracompact Hausdorff space is normal, and thus T_4.

7.6. Partitions of unity

Definition 7.6.1 (Support)

The support of $f : X \to \mathbb{R}$ is the subset of the function's domain solely containing the elements that are not mapped to zero:

$$\operatorname{supp}(f) := \{x \in X : f(x) \neq 0\}$$

Definition 7.6.2 (Partition of unity)

A ***partition of unity*** subordinate to the open cover $\mathcal{U}$ is a family of continuous functions $f_U : X \to [0, 1]$, each associated with a $U \in \mathcal{U}$, such that:
- the support of f_U is contained in U
- each point has a neighborhood on which only finitely many of the f_U are nonzero
- for every $x \in X$:

$$\sum_{U \in \mathcal{U}} f_U(x) = 1$$

Theorem 7.6.3

Given any open cover $\mathcal{U}$ of a paracompact Hausdorff space X, there is a partition of unity subordinate to U.

8. Metrizability

8.1. Metrizability

Theorem 8.1.1

- Every subspace of a metric space is metrizable.
- Countable products of metrizable spaces are metrizable.
- All metric spaces are $T_)5$.
- Metric spaces are first countable.
- Metric spaces are second countable iff they are separable.
- In a metric space, compactness, sequential compactness, and countable compactness are equivalent. Each of these implies second countability.
  - Second countability does not imply any of the compactness properties.

Theorem 8.1.2 ("A. H. Stone")

Every metrizable space is paracompact.

Definition 8.1.3 (G_δ)

A set is $\boldsymbol{G_\delta}$ iff it is a countable intersection of open sets.

- open sets are trivially G_δ
- closed sets in metric spaces are G_δ.
- A single point in a first countable Hausdorff space is G_δ.

Definition 8.1.4 (F_σ)

A set is $\boldsymbol{F_\sigma}$ iff it is a countable union of closed sets.

- closed sets are trivially G_δ.
- open sets in metric spaces are F_σ.

8.2. Total boundedness

Definition 8.2.1 (Totally bounded)

A metric space is ***totally bounded*** iff for each $\varepsilon > 0$ the space is a finite union of sets of diameter at most ε.

Totally bounded spaces are bounded.

Lemma 8.2.2

X is totally bounded iff every sequence contains a Cauchy subsequence.

Lemma 8.2.3

A totally bounded metric space is separable.

Definition 8.2.4 (Complete metric space)

A metric space is ***complete*** iff every Cauchy filter converges.

Theorem 8.2.5

A metric space is compact iff it is complete and totally bounded.

Corollary 8.2.5.1

Subsets of $\mathbb{R}^n$ are compact iff they are closed and bounded.

Theorem 8.2.6 (Lebesgue Number Lemma)

Let $\mathcal{U}$ be an open covering of a compact metric space. Then, there exists a number $\lambda > 0$, called the ***Lebesgue number*** of the cover, s.t. if A is a subset of X with diameter at most λ, then A is contained in some $U \in \mathcal{U}$.

Corollary 8.2.6.1

Every continuous function from a compact metric space to a metric space is uniformly continuous.

8.3. Metrization theorems

Theorem 8.3.1 (Smirnov-Nagata Metrization Theorem)

A necessary and sufficient condition that a topological space be metrizable is that it be a T_3 space with a countable locally finite base.

Theorem 8.3.2 (Urysohn's Metrization Theorem)

A second-countable T_3 space is metrizable.

9. Density and sparsity

Definition 9.1 (Dense set)

A subset A of a topological space X is a set that meets any of the following equivalent conditions:

- $\overline{A} = X$
- $(X - A)^\circ$ is empty
- Every point in X either belongs to A or is a limit point of A
- For every $x \in X$, every neighborhood U of x intersects A
- A intersects every nonempty open subset of X

If $\mathcal{B}$ is a base for X, then the following conditions are equivalent:

- For every $x \in X$, every basic neighborhood $B \in \mathcal{B}$ of x intersects A.
- A intersects every nonempty $B \in \mathcal{B}$

Definition 9.2 (Nowhere dense)

A subset A of a topological space is ***nowhere dense*** in the space iff $\overline{A}^\circ = \varnothing$.

Lemma 9.3

In complete metric spaces and locally compact T_2:

- the intersections of countable collections of dense open sets are dense.
- the countable unions of nowhere dense closed sets are nowhere dense.

9.1. Categories

See 23b Categories

Definition 9.1.1 (First category)

A set is ***first category*** or ***meager*** if it is a countable union of nowhere dense sets.

It is first category in X if it is a countable union of sets that are nowhere dense in X.

Definition 9.1.2 (Second category)

A set is ***second category*** iff it is not first category.

Lemma 9.1.3

Countable unions of first category sets are first category.

Theorem 9.1.4 (Baire Category Theorem)

Complete metric spaces and locally compact T_2 spaces are second category in themselves.

Lemma 9.1.5

The reals are second category. The rationals are first category, while the irrationals are second category.

Theorem 9.1.6

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on a dense subset. Then the set of discontinuities of f is first category.

Corollary 9.1.6.1

There is no function that is continuous at rationals but discontinuous at irrationals.

There is a function that is discontinuous at rationals and continuous at irrationals:

$$f(x) := \begin{cases} 0 : x \text{ irrational} \\ \frac{1}{q} : x = \frac{p}{q} \text{ in lowest terms} \end{cases}$$

Theorem 9.1.7

Let $f_n : \mathbb{R} \to \mathbb{R}$ be a sequence of continuous functions converging pointwise to some function $f(x)$. Then the set of discontinuities of $f(x)$ is first category.

10. Function spaces

10.1. Pointwise convergence

Definition 10.1.1 (Product topology on Y^X)

Let Y be a topological space, and X be a set. Then the product topology on Y^X is the product topology on

$$\prod_{x \in X} Y$$

Definition 10.1.2 (Topology of pointwise convergence)

Let F be a subset of Y^X (the set of all functions $X \to Y$), where Y is a topological space.

The ***topology of pointwise convergence*** is the topology in which a sequence of functions f_n converges to f iff for each $x \in X$ the sequence $f_n(x)$ converges to $f(x)$ in Y. Equivalently, sequences converge iff they converge pointwise.

It is also called the ***point-open topology***, since the set of all

$$S(x, U) := \{f \in Y^X : f(x) \in U\}$$

for each $x \in X$ and U open in Y, forms a subbase for the topology.

This topology is the subspace topology of F from Y^X endowed with the product topology.

10.2. Compact-open topology

See 25 The Compact-Open Topology

Definition 10.2.1 (Evaluation map)

The ***evaluation map***, $E(f, x) : F \times X \to Y$, is defined as

$$E(f, x) := f(x)$$

Lemma 10.2.2

The evaluation map, is:
- continuous in the first argument if F has the point-open topology
- continuous in the second argument if F is a subset of the continuous functions from X to Y

However, the evaluation map is not continuous in both variables together.

Definition 10.2.3 (Compact-open topology)

The ***compact-open topology*** is the topology on F generated by the $\langle K, U \rangle$ where K is compact in X and U is open in Y.

10.2.1. Algebra

Definition 10.2.4 (Ring)

Something that you can do:
- add two elements of the ring
- multiply two elements of the ring
- multiply an element of the ring with a scalar

Definition 10.2.5 (Algebra)

Given rings A and B, if there exists $f : A \to B$, B is a ***algebra*** on A.

Definition 10.2.6 (Subalgebra)

A subset of an algebra that is closed under:
- addition
- multiplication
- scalar multiplication

10.3. Stone-Weierstraß Approximation Theorem

See 27 Stone-Weierstraß Approximation Theorem

Lemma 10.3.1

There exists a sequence $\{p_n(t)\}$ of polynomials that converges uniformly to $\sqrt{t}$ on the unit interval.

Lemma 10.3.2

$$|a+b| + a + b = 2\max(a,b)$$
$$\min(a,b) = a + b - \max(a,b)$$

Definition 10.3.3 ($C^*(X)$)

$C^*(X)$ is the set of all bounded continuous real-valued functions on X. The metric on $C^*(X)$ is the ***supremum metric***:

$$d(f,g) := \sup_{x \in X} |\ f(x) - g(x)\ |$$

$C^*(X)$ is a normed algebra over the real numbers. You can add and multiply its members.

Theorem 10.3.4 (Stone-Weierstraß Approximation Theorem)

Let X be a compact Hausdorff space. Then, any closed subalgebra of $C^*(X)$ that contains constants and separates points must be all of $C^*(X)$.

11. Functional analysis

11.1. Topological vector space

We only care about the fields $F = \mathbb{R}$ and $F = \mathbb{C}$, with their standard topologies.

Definition 11.1.1 (Topological vector space)

A vector space s.t. addition and scalar multiplication are continuous functions.

Definition 11.1.2 (Normed vector space)

A vector space together with a norm $\| \cdot \| : X \to \mathbb{R}$ that satisfies the following properties:

1. **Positive**: $\| \ x \ \| \geq 0$
2. **Definite**: $\| \ x \ \| = 0 \iff x = 0$
3. **Scaling**: $\| \ \alpha x \ \| = |\alpha| \ \|x\|$ for all scalars α
4. **Triangle Inequality**: $\| \ x + y \ \| \leq \|x\| + \|y\|$

$d(x,y) := \| \ x - y \ \|$ is a metric, which also scales. Thus, all normed spaces are metric spaces.

All normed vector spaces are topological vector spaces.

Definition 11.1.3 (Bonoch space)

A complete normed space.

All finite-dimensional normed spaces are Bonoch.

12. Past homework problems

12.1. HW01 Set Basics

1. For all sets A, B, C, prove that $A \cap (B - C) = (A \cap B) - (A \cap C)$.
2. Prove that a function $f : A \to B$ is an injection if and only if $f(X \cap Y) = f(X) \cap f(Y)$ for all subsets X and Y of A.
3. Prove that two equivalence classes of the same equivalence relation on a set are either equal or disjoint.
4. Let R be an equivalence relation on A. Prove that the set of equivalence classes of A under R (denote A/R) is a partition of A. Conversely, given a partition $\mathcal{P}$ of A, prove that there exists a unique equivalence relation R on A such that $\mathcal{P} = A/R$.
5. Prove that, for any set A, there is a bijection from the power set of A, $\mathcal{P}(A)$, to the function set 2^A.
6. Let A and B be partially ordered sets and define a relation $\leq$ on $A \times B$ by $(a,b) \leq (a',b')$ in $A \times B$ iff both $a \leq a'$ in A and $b \leq b'$ in B. (Please excuse the abuse of notation using the same

symbol for all three relations despite the fact that they are certainly not the same sets of ordered pairs!)

a. Prove that this is a partial order on $A \times B$.

b. Give an explicit example showing that $A \times B$ need not be totally ordered even if both A and B are.

7. A totally ordered set S is called ***dense in itself*** if, whenever $x, y \in S$ with $x < y$, there is a $z \in S$ such that $x < z < y$. Prove that two countable totally ordered sets, neither of which has a first or last element, and both of which are dense in themselves, are order-isomorphic.

8. Given $f : A \to B$

a. Prove that the induced map $\overline{f} : \mathcal{P}(A) \to \mathcal{P}(B)$ preserves order (under subsets) and commutes with union.

b. Prove that the induced map $\overline{f^{-1}} : \mathcal{P}(B) \to \mathcal{P}(A)$ preserves order and commutes with union, intersection, and complement.

12.2. HW02 Posets

1. Prove that a poset in which every subset has a supremum is a complete lattice.

2. Let $f : A \to A$ be an order preserving injection of a well-ordered set into itself. Prove that $a \leq f(a)$ for all $a \in A$.

3. Prove that a well-ordered set is not order-isomorphic to any of its initial segments.

4. Prove that there exists an uncountable well-ordered set with a last element Ω such that if $a < \Omega$ then the initial segment of a is countable.

5. For any subset A of a metric space and any real number $\varepsilon > 0$, prove that $U(A, \varepsilon)$ is open.

6. A metric d on M is *bounded* if there is a constant c such that $d(x, y) \leq c$ for all $x, y \in M$. Given a metric d on M, define d^* by $d^*(x, y) = \min(d(x, y), 1)$. Prove that d^* is a bounded metric on M such that d and d^* have the same collection of open sets.

12.3. HW03 Topology

1. How many topologies are there on a 3-element set? How many are distinct under rearrangement of the points?

2. Let $\mathcal{T}, \mathcal{T}'$ be topologies on $\mathcal{X}$ and $\mathcal{X}'$, respectively. Determine (with proof or counterexample) whether the collection $\{U \times U' : U \in \mathcal{T}, U' \in \mathcal{T}'\}$ is a topology on $\mathcal{X} \times \mathcal{X}'$.

3. For a closed set $A \subset X$, prove that $\left(\overline{A^\circ}\right)^\circ = A^\circ$.

> **Note**
>
> We have seen that the interior of a set can be found by closures and complements: $A^\circ = X \setminus \overline{(X \setminus A)}$. (Similarly, closures can be found in terms of interiors and complements.) *Kuratowski's Closure-Complement Theorem* asserts that starting with a set A, there are at most 14 different sets that can be obtained by repeatedly taking closures and complements in some order. Challenge: prove this, and find a set of real numbers that achieves the bound of 14 sets.

4. If U is an open subset of A such that $(A - U)^\circ = \varnothing$, does it follow that $U = A^\circ$? Proof or counterexample.

There is a counterexample.

5. Prove that in a metric space M, $\overline{A} = \{x \in M : d(x, A) = 0\}$.

6. Prove that every closed subset of a metric space is the intersection of a countable family of open sets.

7. If A and B are subsets of a metric space such that $A \cap \overline{B} = \overline{A} \cap B = \varnothing$, prove that there exist disjoint open subsets U and V such that $A \subset U$ and $B \subset V$.

8. A subset A of a topological space is *nowhere dense* in the space if $\overline{A}^\circ = \varnothing$. Prove that a closed subset is nowhere dense if and only if it is the boundary of some open set.

12.4. HW04 Using the Basics

1. Let $A \subset X$ and $B \subset Y$. In the product space $X \times Y$ prove that $(A \times B)^\circ = A^\circ \times B^\circ$. (Clearly this can extend to any finite product by induction.)

2. Let $X = \prod X_\lambda$ where X_λ, $\lambda \in \Lambda$ be any product of topological spaces, with the product topology. Let $A = \prod A_\lambda$ where each $A_\lambda \subset X_\lambda$. prove that $\overline{A} = \prod \overline{A_\lambda}$.

3. A topological space is ***separable*** if it has a countable dense subset (that is, a countable set whose closure is the entire space). Prove that a metric topology is separable iff it is second countable.

4. Prove that every second countable space is separable.

5. Prove that the Sorgenfrey line is separable.

> **Note**
>
> The Sorgenfrey line thus provides an example of a space that is separable and first countable, but still not second countable.

6. Prove that an open subset of a separable space is separable; given an example of a separable space with a subspace that is not separable.

7. A collection of functions $f_\lambda : X \to X_\lambda$ *separates points from closed sets* if, given closed $A \subset X$ and a

point $x \notin A$ then there is some λ such that $f_\lambda(x) \notin \overline{f_\lambda(A)}$.

If $\{f_\lambda\}$ is a collection of continuous functions that separates points from closed sets, prove that X has the topology induced by the fs.

8. Prove that a countable product of first countable spaces is first countable.

12.5. HW05 Filters and Separation

1. Under what conditions is a filter equal to the intersection of all the ultrafilters that contain it?

A filter is always equal to the intersection of all the ultrafilters containing it.

2. Prove that a filter which is contained in a unique ultrafilter is an ultrafilter.
3. A filter $\mathcal{F}$ in a metric space is a ***Cauchy filter*** iff for every $\varepsilon > 0$ there is an $F \in \mathcal{F}$ with diameter less than ε. (The diameter of a set F is $\sup\{d(x, y) : x, y \in F\}$.) Prove that if $\mathcal{F}$ converges in the metric topology then it is a Cauchy filter.
4. Prove that every Cauchy filter in $\mathbb{R}$ converges.
5. In a T_1 space, prove that x is an accumulation point of a subset $A \subset X$ iff every neighborhood of x contains infinitely many points of A.
6. Let $f_n : X \to Y$ be a sequence of continuous functions from a topological space X to a metric space Y that *converges uniformly* to a function f. That is, for every $\varepsilon > 0$ there is an integer N such that $d(f_n(x), f(x)) < \varepsilon$ for all $n \geq N$ and all $x \in X$. Prove that the limit function f is also continuous.
7. (Skipped) Let $\mathcal{U}$ be an open covering of a normal space X such that every point in x belongs to only finitely many of the $U \in \mathcal{U}$. Prove that there exists another open covering $\mathcal{V} = \{V_U : U \in \mathcal{U}\}$ of X such that $\overline{V_U} \subset U$ for every $U \in \mathcal{U}$.
8. Prove that every open covering of the Sorgenfrey line contains a countable subcovering.
9. Prove that every open subset of the reals is the union of a countable family of pairwise disjoint open intervals.

12.6. HW06 Assorted

1. Let $\mathcal{F}$ be a filter base in a topological space X. Prove that $\mathcal{F}$ converges to a point $x_0 \in X$ iff every ultrafilter containing $\mathcal{F}$ converges to x_0.
2. Prove that a topological space X is Hausdorff iff for every topological space Y and every pair of continuous functions f and g from Y to X, the set $\{y \in Y : f(y) = g(y)\}$ is a closed subset of Y.
3. Prove that a topological space X is completely normal iff every pair of subsets A and B s.t. $\overline{A} \cap B = \varnothing = A \cap \overline{B}$ can be separated by neighborhoods.
4. Let X be a set and Y a metric space with metric $d(x, y)$. Let $B(Y^X)$ denote the set of all bounded functions from X to Y (that is, $\sup(\{d(f(x_1), f(x_2))\})$ is finite). Prove that $d'(f, g) := \sup(\{d(f(x), g(x)) : x \in X\})$ is a metric on $B(Y^X)$.
5. Let X be a topological space. Let $C \subset B(Y^X)$ be the set of all bounded continuous functions. Prove that C is a closed subset of $B(Y^X)$.
6. (Extra credit) Let $\{U_\lambda\}_{\lambda \in \Lambda}$ be a locally finite open covering of a normal space X. Prove that there is a collection of continuous functions $\{f_\lambda : X \to [0, 1]\}_{\lambda \in \Lambda}$ such that $f_\lambda(X - U_\lambda) = 0$ and $\sum_{\lambda \in \Lambda} f_\lambda(x) = 1$ for all $x \in X$. (Note that each sum is actually a finite sum, since by the local finiteness of the Us only a finite number of the summands are nonzero. Such a collection is called a ***partition of unity subordinate to*** $\{U_\lambda\}$).

12.7. HW07 More Assorted

1. A ***zero set*** in a topological space is the set of zeroes for some continuous real-valued function on X. Prove that in a normal space, a necessary and sufficient condition for a set to be a zero set is that it be closed and a countable intersection of open sets.
2. A subset A of a topological space is ***nowhere dense*** in X iff every nonempty open set $U \subset X$ contains a nonempty open set V s.t. $V \cap A = \varnothing$. Let $A \subset B \subset C \subset X$ and assume that B is nowhere dense in C. Show that A is nowhere dense in C and B is nowhere dense in X.
3. Prove the following are equivalent:
   1. A is nowhere dense in X.
   2. $X - \overline{A}$ is dense in X.
   3. $\overline{A}$ is nowhere dense in X.
4. Prove that a connected Tychonoff space with more than one point has uncountably many points.
5. Prove that an open interval (a, b) is not homeomorphic to a half-open interval $[c, d)$, both subspaces of the line.
6. Prove that $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^n$ for any $n > 1$.
7. Prove that every continuous map of the closed unit interval to itself has a fixed point. Decide whether the result is true for the open unit interval. Decide whether the result is true for bounded continuous functions from $\mathbb{R}$ to itself.
8. Prove that there is no continuous injective map from the circumference of the unit circle (with the induced topology from the plane) into $\mathbb{R}$.

9. (Extra credit) A metric space is ***complete*** iff every Cauchy filter converges. Prove that, in a complete metric space, the intersection of a countable family of dense open subsets is dense.

13. Summary sections

13.1. Closure

Definition 2.3.1 (Closure)

Given $A \subset X$ where X is a topological space, the ***closure*** of A, denoted $\overline{A}$ or cl A, is the intersection of all closed subsets containing A.

Theorem 2.4.4 (Neighborhood characterization of closure)

Let $x \in X$, $A \subset X$. Then $x \in \overline{A}$ iff every neighborhood of x meets A.

Corollary 2.4.10.1 (Accumulation point characterization of closure)

$\overline{A} = A \cup A'$.

Theorem 3.1.5

In a first countable space, $x \in \overline{A}$ iff there is a sequence of points in A that converge to x.

Lemma 3.2.4 (Net characterization of closure)

If A is a subset of a topological space, $x_0 \in \overline{A}$ iff there exists a net in A converging to x_0.

Lemma 3.3.12 (Filter base characterization of closure)

$x \in \overline{A}$ iff there is a filter base in A converging to x.

13.2. Continuity

Definition 2.5.1 (Continuous function)

$f : X \to Y$ is a continuous function at $x_0 \in X$ iff $f^{-1}(N)$ is a neighborhood of x_0 for every neighborhood N of $f(x_0)$.

Equivalently, it is continuous iff every neighborhood of $f(x)$ pulls back (by f^{-1}) to a neighborhood of x.

f is continuous if it is continuous at all $x_0 \in X$.

Theorem 2.1.2 (Metric characterization of continuity)

$f : M \to N$ is continuous at $x \in M$ iff for any $\varepsilon > 0$, there exists $\delta > 0$ s.t. $d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon$.

Theorem 2.1.4 (Epsilon neighborhood definition of continuity)

$f : M \to N$ is continuous at $x \in M$ iff for every open set $V \subset N$ that contains $f(x)$, there is an open set $U \subset M$ containing x s.t. $f(U) \subset V$.

Lemma 3.2.5 (Net characterization of continuity)

$f : X \to Y$ is continuous at x_0 iff for each net $x_\alpha \to x_0$, $f(x_\alpha) \to f(x_0)$.

Lemma 3.3.11 (Filter characterization of continuity)

Let $f : X \to Y$, and $x_0 \in X$ where N is the set of neighborhoods of x_0. Then, f is continuous at x_0 iff $N \to f(x_0)$.

13.3. Common/useful topologies

Definition 13.3.1 (Standard topology on $\mathbb{R}$)

A set is open iff it is the union of open intervals.

Definition 13.3.2 (Discrete topology)

All sets are open:

$$\mathcal{T} = \mathcal{P}(X)$$

This is equivalent to the topology given by the discrete metric.

Definition 13.3.3 (Indiscrete topology)

Only the entire topological space and $\varnothing$ are open.

This is a metrizable space given by the indiscrete pseudometric.

Definition 13.3.4 (Metric topology)

The topology generated by the base consisting of all open balls $U(x, \varepsilon)$.

Definition 13.3.5 (Cofinite topology)

All subsets whose complements are finite are open. $\varnothing$ is also open.

Definition 13.3.6 (Sorgenfrey line)

The topology generated by the base consisting of all half-open intervals $[a, b)$.

Definition 13.3.7 (Excluded-point topology)

Choose a point x in a set X. All sets not containing x, along with the entire space X, are open.

Definition 13.3.8 (Order topology / interval topology)

On a well-ordered set X, a set is open iff it is the union of open intervals.

Definition 13.3.9 (Slotted plane)

The topology on $\mathbb{R}^2$ generated by the neighborhood base at each point x of the set of disks missing their central diameter but including x itself.

Definition 13.3.10 (Moore plane)

The topology on the closed upper half-plane generated by the neighborhood base of:

- For points above the x-axis, the set of open disks about the point which do not touch the x-axis
- For points on the x-axis, the set of open disks tangent to the x-asix at the point, along with the point itself

Definition 13.3.11 (Ω topology)

The ***open Ω topology*** is the order topology on the well-ordered set $[0, \Omega)$ (the least uncountable well-ordered set).

The ***closed Ω topology*** is the order topology on the well-ordered set $[0, \Omega]$ (the least uncountable well-ordered set plus Ω itself).

14. Index

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